

A comparative analysis of Ottoman-turkish Makams and Hindustani Ragas

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Ottoman-Turkish Makam

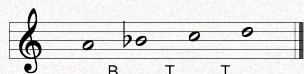
- Thousand years old tradition related to Arabic maqams
- Each makam specifies a unique intervallic structure (*cinsler*) and melodic development (*seyir*)
- Uses 53-TET system
- Most makams are built of a tetrachord built atop a pentachord, or vice versa



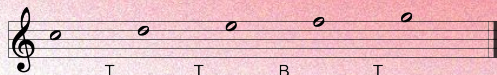
Çârgâh Dörtlüsü



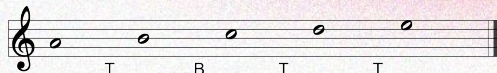
Bûselik Dörtlüsü



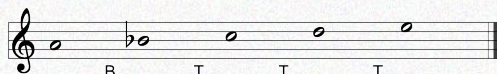
Kürdi Dörtlüsü



Çârgâh Beşlisi



Bûselik Beşlisi



Kürdi Beşlisi

North Indian (Hindustani) Ragas

- Hindustani classical music was influenced by Persian and Islamic culture
- Each raga consists of an array of melodic structures with musical motifs
- Microtonality is detected in ornamentations
- Each raga has a tonic pitch that is the base frequency that serves as a performance reference

Mârvâ

Sa Re:b Ga Ma:# Pa Dha Ni Sa

Bhairavi

Sa Re:b Ga:b Ma Pa Dha:b Ni:b Sa

Āsāvāri

Sa Re Ga:b Ma Pa Dha:b Ni:b Sa

Research question and background

“Can symbolic melodic similarity metrics identify structural correspondences between Hindustani Ragas and Ottoman-Turkish Makams, and what do these correspondences reveal about the shared melodic grammar of these multimodal traditions?”

Related work

- “A study of the Raga Zeelaf and its relationship with Arabian traditional music”, A. Bhattacharya and D. K. Das, The Journal of Acoustical Society of India, vol. 50, no. 3–4, pp. 121–127, 2023.
- “Aspects generating variety in eastern melodic multimodality”, M. Skoulios, in Minutes of the 2017 Arel Symposium, Istanbul, Turkey, December 2018
- “MelodyShape at MIREX 2014 symbolic melodic similarity”, J. Urbano, in 10th Music Information Retrieval Evaluation eXchange (MIREX 2014), Taipei, Taiwan, October 2014
- “A comparison of symbolic similarity measures for finding occurrences of melodic segments” P. Janssen, P. Van Kranenburg, and A. Volk



Methodology

01

Data Preprocess

02

Metrics

03

Analysis

- Ragas: SANGEET dataset, 116 compositions of 3 Ragas: Bhairav (42), Todi (39), and Poorvi (35)
- Turkish Makams: SymbTr dataset, 1,700 compositions of 155 Makams and 100 Usuls
- Turkish makams had the standard XML form
- For hindustani ragas we had to create a custom parser

Hindustani Ragas custom parser based on SWARALIPI framework*

नि ध प म | ग रे ग म | नि ध म ऽ | प म ग रे
० ३ x २

(a)

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22 </ROW>
23 .....
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(b)

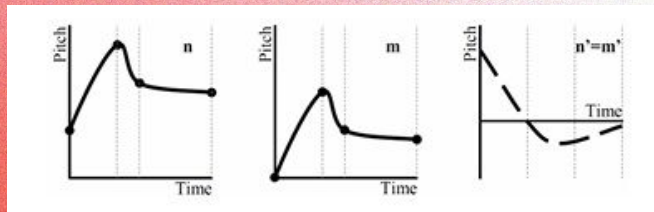
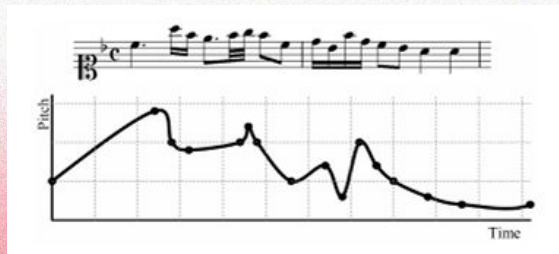
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(c)

Fig. 1: Sheet part of the XML file (b) and (c) depicting part of the original music sheet (a).

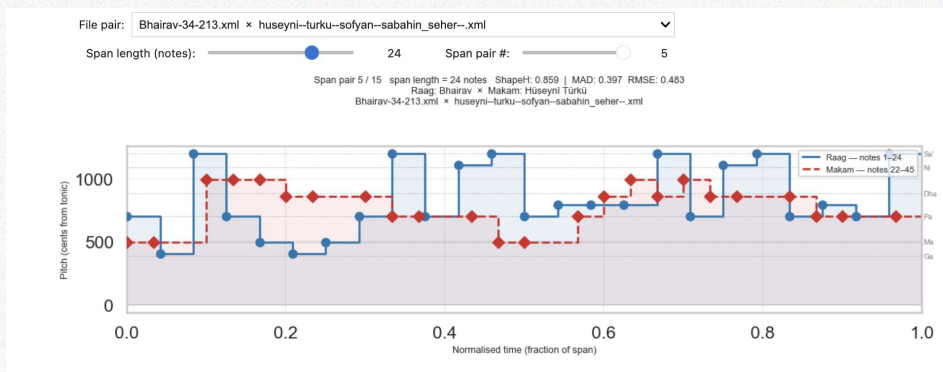
*Misra, Chandan. (2016). SWARALIPI: A FRAMEWORK FOR TRANSCRIBING AND RENDERING INDIC MUSIC SHEET.

02 Metrics



- The similarity between two melodies was computed with the sequence alignment algorithm MelodyShape
- Melodies are compared by fitting polynomial curves to pitch and timing contours
- Transposition, Time-scale and position invariance
- Similarity metrics: ShapeH (shape-based) and Time (continuous curve-based)
- Drawback: Reducing a continuous value into discrete points
- Transition trajectory information was lost

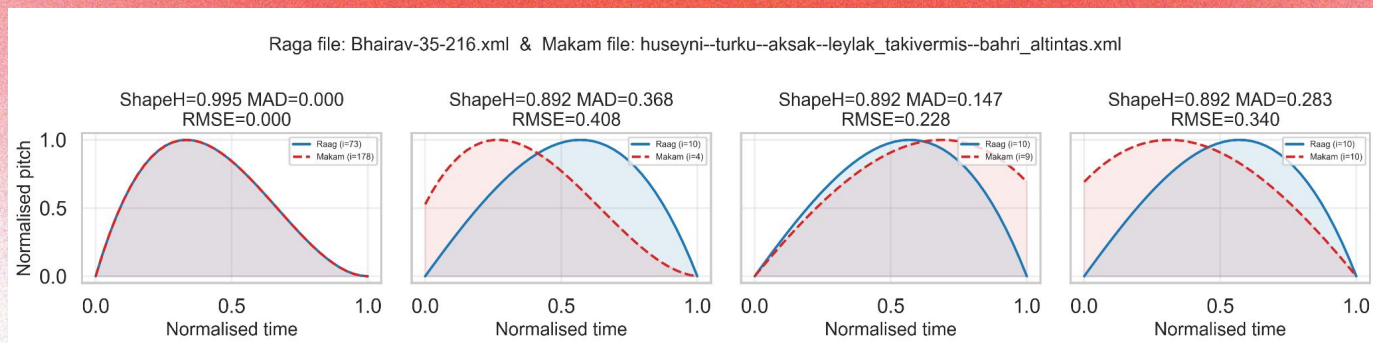
03 Analysis



- Compositions were organized and compared according to their Raga/Makam
- Similarity score was computed pairwise (ShapeH)
- For most similar Raga and Makams Time metric was computed (ShapeTime system)
- Top results were displayed

Results

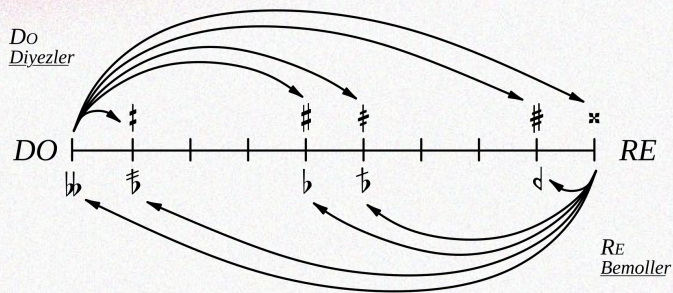
- Raga Bhairav showed the highest score of similarity to Makam melodic phrases
- Hüseyni Türkü Makam had the highest score of similarity to Raga Bhairav
- Raga Todi had the lowest score of similarity



Results

The melodic similarity between Bhairav and several Turkish Makams can be explained by shared structural characteristics at both the intervallic and tetrachordal level

- Use of microtonalities
- presence of asymmetric tetrachords



Conclusion

- Ottoman-Turkish Makams and North Indian (Hindustani) Ragas are known for their richness and organization using a variety of modes
- For comparison MusicXML datasets and MelodicShape algorithm were used
- Raga Bhairav's melodic curves consistently matched with several Turkish Makams
- This can be attributed to shared structural characteristics

Limitations/Future work

- MusicXML notation cannot represent sufficiently transition trajectories
- Hindustani dataset was limited to 3 Ragas
- Combining symbolic analysis with audio-based approaches would allow better representation
- Perform more detailed analysis

Thank you